

Lixin Liu (刘李鑫)

Zhongguancun Laboratory | Beijing, China | <https://llxsd.github.io/>
liulixinsd@gmail.com

Research Profile

My research focuses on emerging architectural challenges in Low-Earth-Orbit (LEO) satellite networks as they scale to serve diverse users and support multiple operators.

Education

Tsinghua University - Cyberspace Security - Doctorate	Sep 2022 - Jun 2026
Tsinghua University - Cyberspace Security - Master	Sep 2019 - Jun 2022
Shandong University - Software Engineering - Bachelor	Sep 2015 - Jun 2019

Working Experiences

Zhongguancun Laboratory - Assistant Researcher	Jul 2026 - Present
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Honors & Awards

- Jun, 2026 **Beijing Outstanding Graduate**
- Jun, 2026 Outstanding Graduate, Institute for Network Sciences and Cyberspace, Tsinghua University
- Oct, 2025 Top Grade Scholarship, Zhongguancun Laboratory
- Oct, 2025 Jiang Nanxiang Scholarship, Tsinghua University
- Oct, 2024 **National Scholarship**
- Apr, 2024 **USENIX NSDI 2024 Outstanding Paper Award**

Publications

- [[USENIX NSDI 2024](#)] [Lixin Liu](#), Yuanjie Li, Hewu Li, Jiabo Yang, Wei Liu, Jingyi Lan, Yufeng Wang, Jiarui Li, Jianping Wu, Qian Wu, Jun Liu, Zeqi Lai, Democratizing Direct-to-Cell Low Earth Orbit Satellite Networks. **Outstanding Paper Award**
- [[IEEE INFOCOM 2026](#)] [Lixin Liu*](#), Jingyi Lan*, Bijia You, Yuanjie Li, Hewu Li, Qian Wu, Zeqi Lai, The Invisible Hand: Regaining Control of Service Quality from Outsourced Satellite Networks.
- [[ACM MobiCom 2024](#)] Yuanjie Li, [Lixin Liu](#), Hewu Li, Wei Liu, Yimei Chen, Wei Zhao, Jianping Wu, Qian Wu, Jun Liu, Zeqi Lai, Stable Hierarchical Routing for Operational LEO Networks.
- [[NINeS 2026](#)] [Lixin Liu](#), Jinyao Zhang, Bijia You, Yimei Chen, Jiabo Yang, Yuanjie Li, Hewu Li, Qian Wu, Zeqi Lai, Jun Liu, CrowdLink: Unlocking Idle LEO Network Capacity with User Terminals.
- [[ACM HotNets 2024](#)] Yimei Chen*, [Lixin Liu*](#), Yuanjie Li, Hewu Li, Qian Wu, Jun Liu, Zeqi Lai, Unraveling Physical Space Limits for LEO Network Scalability.
- [[IEEE/ACM IWQoS 2022](#)] [Lixin Liu](#), Hewu Li, Yuanjie Li, Zeqi Lai, Yangtao Deng, Yimei Chen, Wei Liu, Qian Wu, Geographic Low-Earth-Orbit Networking without QoS Bottlenecks from Infrastructure Mobility.
- [[IEEE Internet Computing 2024](#)] [Lixin Liu](#), Wei Liu, Yuanjie Li, Hewu Li, Enabling 6G and Beyond Network Functions From Space: Challenges and Opportunities.
- [[ACM HotNets 2021](#)] Yuanjie Li, Hewu Li, [Lixin Liu](#), Wei Liu, Jiayi Liu, Jianping Wu, Qian Wu, Jun Liu, Zeqi Lai, "Internet in Space" for Terrestrial Users via Cyber-Physical Convergence.
- [[ACM MobiCom 2023](#)] Yuanjie Li, Hewu Li, Wei Liu, [Lixin Liu](#), Wei Zhao, Yimei Chen, Jianping Wu, Qian Wu, Jun Liu, Zeqi Lai, Han Qiu, A Networking Perspective on Starlink's Self-Driving LEO Mega-Constellation.
- [[ACM SIGCOMM 2022](#)] Yuanjie Li, Hewu Li, Wei Liu, [Lixin Liu](#), Yimei Chen, Jianping Wu, Qian Wu, Jun Liu, Zeqi Lai, A Case for Stateless Mobile Core Network Functions in Space.

11. [\[ACM SIGCOMM 2025\]](#) Yuanjie Li, Yimei Chen, Jiabo Yang, Jinyao Zhang, Bowen Sun, [Lixin Liu](#), Hewu Li, Jianping Wu, Zeqi Lai, Qian Wu, Jun Liu, Small-scale LEO Satellite Networking for Global-scale Demands.

12. [\[ACM SIGCOMM 2025\]](#) Wei Liu, Yuanjie Li, Jingyi Lan, Hewu Li, Yimei Chen, [Lixin Liu](#), Jiabo Yang, Xi Long, Li Ouyang, Minghao Tang, Jianping Wu, Qian Wu, Jun Liu, Zeqi Lai, Direct-to-Cell Satellite Network without Satellite Navigation.

13. [\[USENIX NSDI 2023\]](#) Zeqi Lai, Hewu Li, Yangtao Deng, Qian Wu, Jun Liu, Yuanjie Li, Jihao Li, [Lixin Liu](#), Weisen Liu, Jianping Wu, StarryNet: Empowering Researchers to Evaluate Futuristic Integrated Space and Terrestrial Networks.

Patents

1. 李元杰, [刘李鑫](#), 李贺武, 刘威, 陈怡梅, 吴茜, 刘君, 赖泽祺, 陆璐, 郑韶雯, 《星地融合互联网的稳定层次化路由方法、装置及电子设备》, CN119109855B.

2. 李元杰, [刘李鑫](#), 李贺武, 刘威, 蓝静怡, 汪宇峰, 吴茜, 刘君, 赖泽祺, 《直连蜂窝卫星网络的自助服务方法及装置》, CN118102261A.

3. 李元杰, [刘李鑫](#), 蓝静怡, 李贺武, 吴茜, 刘君, 赖泽祺, 《多租户直连蜂窝卫星网络的切换方法、装置以及电子设备》, CN120301494A.

4. 李贺武, [刘李鑫](#), 刘君, 吴茜, 李基豪, 《基于空间位置信息进行网络层寻址的路由方法和路由设备》, CN112865852B.

5. 李元杰, 李贺武, 刘威, [刘李鑫](#), 陈怡梅, 吴茜, 刘君, 赖泽祺, 《网元状态管理方法、装置、电子设备及存储介质》, CN116032344A.

6. 李元杰, 李贺武, 刘嘉懿, 刘威, [刘李鑫](#), 吴茜, 刘君, 赖泽祺, 《一种星地协同组网方法及装置》, CN114039644B.

7. 李元杰, 李贺武, 赵玮, 陈怡梅, 刘威, [刘李鑫](#), 吴茜, 刘君, 赖泽祺, 《卫星自动变轨检测方法、装置、电子设备及存储介质》, CN116896404A.

8. Yuanjie Li, Hewu Li, Jiayi Liu, Wei Liu, [Lixin Liu](#), Qian Wu, Jun Liu, Zeqi Lai, "Method and Device for Constructing Integrated Space-Terrestrial Network", USA, No. US12407407B2.

Professional Service

- Reviewer, IEEE/ACM Transactions on Networking
- Reviewer, IEEE Transactions on Mobile Computing
- Reviewer, Computer Networks
- Artifact Evaluation Committee Member, ACM MobiCom 2024
- Student Volunteer, ACM SIGCOMM 2019

Community Engagement

- Guest Contributor, APNIC Blog, "**The Invisible Hand: Regaining Control of Service Quality from Outsourced Satellite Networks**", May 2026.
- Invited Speaker, Huang Danian Chaspark "Best Paper Deep Dive, Reading Top Meeting" Series, "**Democratizing Direct-to-Cell Low Earth Orbit Satellite Networks**", Nov 2025.
- Speaker, IETF 121 Satellite Networking Side Meeting, "**Stable Hierarchical Addressing and Routing for Operational Satellite Networks**", Nov 2024.
- Participant, IETF 116 Satellite Networking Side Meeting, "**Problems and Requirements of Addressing in Integrated Space-Terrestrial Network (ISTN)**", Mar 2023.